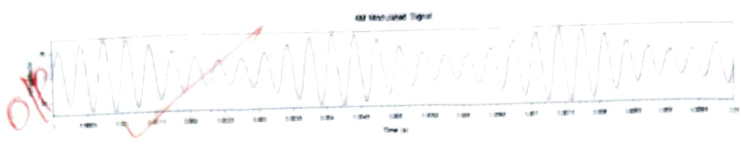
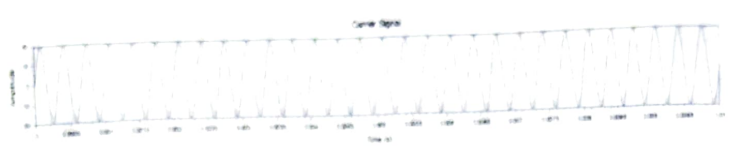


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20/07/2026

EXPERIMENT 1 AMPLITUDE MODULATION

AIM:-

To stimulate amplitude modulation (AM) and modulation index and Bandwidth Analysis

APPARATUS REQUIRED:-

scilab software

CODE:-

```
Em = 10;
Fm = 300;
Ec = 20;
Fc = 3000;
Fs = 30000;

t = 0:1/Fs:3/Fm;

am = Em * sin(2 * %pi * Fm * t);
subplot(3,1,1);
plot(t, am);
title("Message signal");

c = Ec * sin(2 * %pi * Fc * t);
subplot(3,1,2);
plot(t, c);
title("Carrier signal");

eam = (Ec + am) * cos(2 * %pi * Fc * t);
subplot(3,1,3);
plot(t, eam);
title("Am waves");
```

Sno	Description	Theory		Practical		
		Amplitude	Frequency	Amplitude	Time	Frequency
		Theory	Practical	Theory	Time	Practical
1.	Message signal	10	300	10	$T_m = \frac{1}{300}$ $= 0.00334$ $= 0.00417 - 0.00083$ $= 0.00334s$	$f_m = \frac{1}{T_m}$ $= \frac{1}{0.00334}$ ≈ 300
2.	carrier signal	20	3000	20	$T_c = \frac{1}{3000}$ $= 0.000333$ $T = x_2 - x_1$ $\Rightarrow 0.00133 - 0.00100$ $\Rightarrow 0.00033s$	$F_m = \frac{1}{T_m}$ ≈ 3000
3.	Am modulator signal	—	—	$E_{max} = 30$ $E_{min} = 10$	T_{AM} $= 0.000333$ $T = x_2 - x_1$ $\Rightarrow 0.00133 - 0.00100$ $= 0.00033s$	$F_m = \frac{1}{T_m}$ ≈ 3000

CALCULATION:-

Theoretical;

$$M_a = \frac{E_m}{E_c} \Rightarrow \frac{10}{20} \Rightarrow \frac{1}{2} \Rightarrow 0.5$$

Practical;

$$M_a = \frac{E_{max} - E_{min}}{E_{max} + E_{min}} \Rightarrow \frac{20 - 10}{20 + 10} \Rightarrow \frac{10}{30} \Rightarrow 0.33$$

$$\therefore (M_a)_{Theoretical} = (M_a)_{Practical}$$

Scilab program & Implementation (20)	Waveforms & tabulation (20)	Calculation and result (20)	Viva-voce (20)	Result/Observation (20)
20	20	20	15	$\frac{20}{95}$

RESULT:-

AM waves are successfully generated thus the modulated index is calculated.